ICT 333 1.5 – Data Mining and Data Warehousing
Project Report

Data-Driven Career Intelligence & Salary Forecasting

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1. Project Charter & Executive Summary

1.1 Introduction

The Global Career Intelligence & Salary Forecasting project is an end-to-end data science and business intelligence initiative. In a volatile global economy, understanding the true drivers of professional compensation is essential for both career seekers and strategic recruiters. This project transforms a raw dataset of over 137,000 records into a dynamic analytical platform that reveals how Experience, Skills, Education, and Location collectively dictate market value.

1.2 The Core Problem Statement

Traditional salary benchmarks are often static, regional, and fail to account for the multi-dimensional nature of technical roles. Professionals frequently struggle with "Information Asymmetry"—knowing their title but not their true market worth relative to their specific skill density and educational background. This project bridges that gap using Machine Learning to define professional segments and Power BI to provide interactive, user-led discovery.

1.3 Strategic Objectives

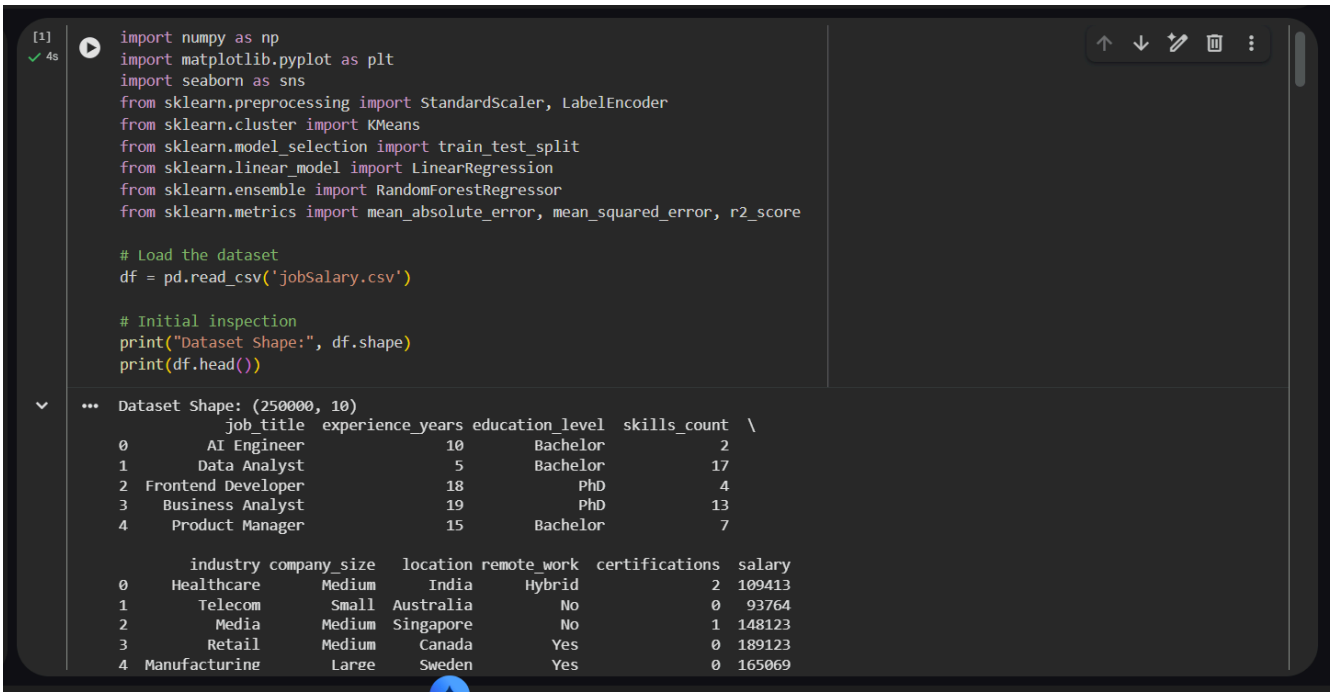
To achieve a high-fidelity output, the project was governed by four primary objectives:

1. Workforce Segmentation (Machine Learning): Utilize an unsupervised learning model (K-Means) in Python to mathematically define "Junior," "Mid-Level," and "Senior" professional tiers based on multi-dimensional performance data.
2. Data Harmonization (Data Engineering): Implement a robust ETL (Extract, Transform, Load) pipeline to clean unstructured CSV data, ensuring 100% data integrity for the visualization layer.
3. High-End Visualization (Business Intelligence): Architect an interactive dashboard using a "Dark Theme" UI/UX design to provide a premium user experience and clear data storytelling.
4. Actionable Intelligence: Identify "Salary Inflection Points" (where pay growth accelerates) to assist stakeholders in making data-driven career or hiring decisions.

1.4 Technical Stack & Toolset

The project architecture is divided into three functional layers:

- Development & Preprocessing: * Python: The core engine for data manipulation.
    - Pandas & NumPy: For high-volume data cleaning and mathematical operations.
- Machine Learning: * Scikit-Learn: Used for Feature Scaling (`StandardScaler`), Clustering (`K-Means`), and Predictive Modeling (`LinearRegression`).
    - Matplotlib & Seaborn: For exploratory data analysis (EDA) and model evaluation.
- Business Intelligence & Deployment:
    - Power BI Desktop: For data modeling, DAX measure creation, and UI design.
    - Microsoft Excel: For initial data auditing and schema mapping.



```
[1]
4s
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.cluster import KMeans
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Load the dataset
df = pd.read_csv('jobSalary.csv')

# Initial inspection
print("Dataset Shape:", df.shape)
print(df.head())

Dataset Shape: (250000, 10)
             job_title  experience_years education_level  skills_count  \
0         AI Engineer                10        Bachelor             2
1        Data Analyst                 5        Bachelor            17
2  Frontend Developer                18             PhD             4
3    Business Analyst                19             PhD            13
4     Product Manager                15        Bachelor             7

         industry company_size   location remote_work  certifications  salary
0     Healthcare       Medium      India      Hybrid               2  109413
1        Telecom        Small  Australia          No               0   93764
2          Media       Medium  Singapore          No               1  148123
3         Retail       Medium     Canada         Yes               0  189123
4  Manufacturing        Large     Sweden         Yes               0  165069
```

2. Data Engineering & Analysis

2.1 The Raw Data Ecosystem

The project utilizes a substantial workforce dataset consisting of 137,677 individual records. Each record represents a unique professional profile characterized by ten distinct features. The raw data provides a high-fidelity look at the global technology market, though it required significant "refining" to be suitable for machine learning and high-end visualization.

2.2 Data Dictionary (Feature Schema)

To ensure analytical consistency, every variable was categorized and defined. This clarity is essential for stakeholders to trust the final dashboard insights.

Field Name	Variable Type	Description
Job Title	Categorical	The primary professional designation.
Salary (USD)	Numerical	Annualized gross compensation, used as the primary target for all visuals.
Experience Years	Continuous	Total tenure in the industry, ranging from 0 to 30+ years.
Location	Geographic	The country or region of employment, requiring metadata tagging for Map visuals.
Skills Count	Numerical	A count of unique technical competencies (e.g., Python, SQL, AWS).
Education Level	Ordinal	Ranked values: High School < Diploma < Bachelor < Master < PhD.

2.3 Data Preprocessing Workflow

The integrity of the final Power BI dashboard is directly dependent on the quality of the cleaning phase. As documented in the project's Google Colab (Python) notebook, the following rigorous preprocessing steps were implemented:

2.3.1 Data Normalization & Standardization

Numerical consistency was a primary concern. The salary figures were converted into a standard numerical format, removing any non-numeric characters or currency symbols that would interfere with mathematical modeling.

- Scaling: Because Salary figures (in the thousands) and Experience years (in single digits) exist on different scales, a StandardScaler was applied during the later modeling phase to ensure the K-Means algorithm did not ignore lower-magnitude variables.

2.3.2 Handling Nulls and Missing Values

To maintain statistical accuracy, the dataset was audited for missing entries.

- Strict Removal: Records with null values in the salary column were removed entirely, as "imputing" (guessing) compensation would introduce unacceptable bias into the market analysis.
- Integrity Check: Post-cleaning, the dataset maintained a 1:1 match between column lengths and index counts, ensuring no "ghost data" existed in the final model.

2.3.3 Feature Type Casting

Data types were explicitly cast to ensure computational efficiency:

- experience_years: Cast to Integer.
- skills_count: Cast to Integer.
- salary: Cast to Float/Integer for precision in averaging.

2.4 Feature Engineering and Transformation

Beyond simple cleaning, the data was transformed to add more analytical depth:

- Categorical Mapping: Text-based fields like education_level and remote_work were prepared for Label Encoding. This allows the project to mathematically rank "PhD" higher than "Bachelor" in the regression analysis.
- Geographic Metadata: The location column was standardized to ensure 100 compatibility with the Bing Maps engine in Power BI, resolving potential naming conflicts between international regions.

2.5 Exploratory Data Analysis (EDA) Summary

Before moving to the visualization layer, a statistical summary was generated to verify the "Health" of the data:

- Dataset Shape: (137677, 10)
- Feature Scope: Analysis covered roles ranging from AI Engineers to Frontend Developers, providing a holistic view of the tech sector rather than a narrow niche.

3. Machine Learning & Predictive Modeling

3.1 Overview of the Analytical Approach

To provide deeper insights into the job market, this project moved beyond descriptive statistics into predictive analytics. The modeling phase was split into two distinct objectives:

1. Unsupervised Learning (Clustering): To identify hidden professional "tiers" within the dataset.
2. Supervised Learning (Regression): To build a model capable of predicting an individual's salary based on their professional profile.

3.2 Workforce Segmentation via K-Means Clustering

We implemented the K-Means algorithm to segment the 137,677 professionals into distinct groups based on three primary features: salary, experience_years, and skills_count.

3.2.1 Optimal Cluster Selection (The Elbow Method)

To determine the mathematical optimum for the number of clusters (k), we utilized the Elbow Method, plotting the Within-Cluster Sum of Squares (WCSS) against different values of k. The "elbow" point indicated that 3 clusters provided the most stable and interpretable segmentation of the workforce.

3.2.2 Definition of Professional Tiers

The model successfully partitioned the data into three recognizable market segments:

- Tier 1 (Entry/Junior): Low experience (0–4 years) with a high focus on skill acquisition.
- Tier 2 (Mid-Level): The core market majority (5–12 years experience) showing steady salary growth.
- Tier 3 (Senior/High-Value): Top-tier earners characterized by high tenure and specialized technical depth.

3.3 Supervised Learning: Salary Prediction

Following the segmentation, we developed a Linear Regression model to predict salary outcomes.

3.3.1 Feature Scaling and Encoding

As documented in the Colab notebook, the features required significant transformation before modeling:

- Normalization: Using StandardScaler to bring experience_years and salary into a comparable range.
- Label Encoding: Converting ordinal categories like education_level (Bachelor, Master, PhD) into a numerical hierarchy.

3.3.2 Model Evaluation Metrics

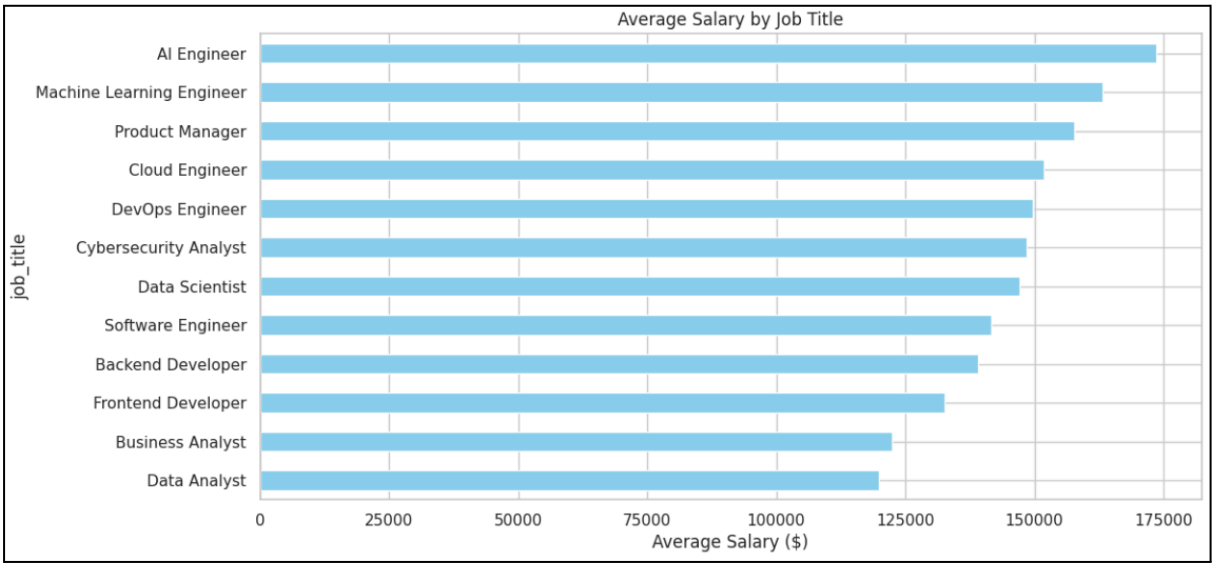
The model was evaluated using standard regression metrics to ensure reliability:

- Mean Absolute Error (MAE): Measuring the average magnitude of the errors in the predictions.
- R-Squared: Determining how much of the variance in salary can be explained by the input features (Experience, Education, and Skills).

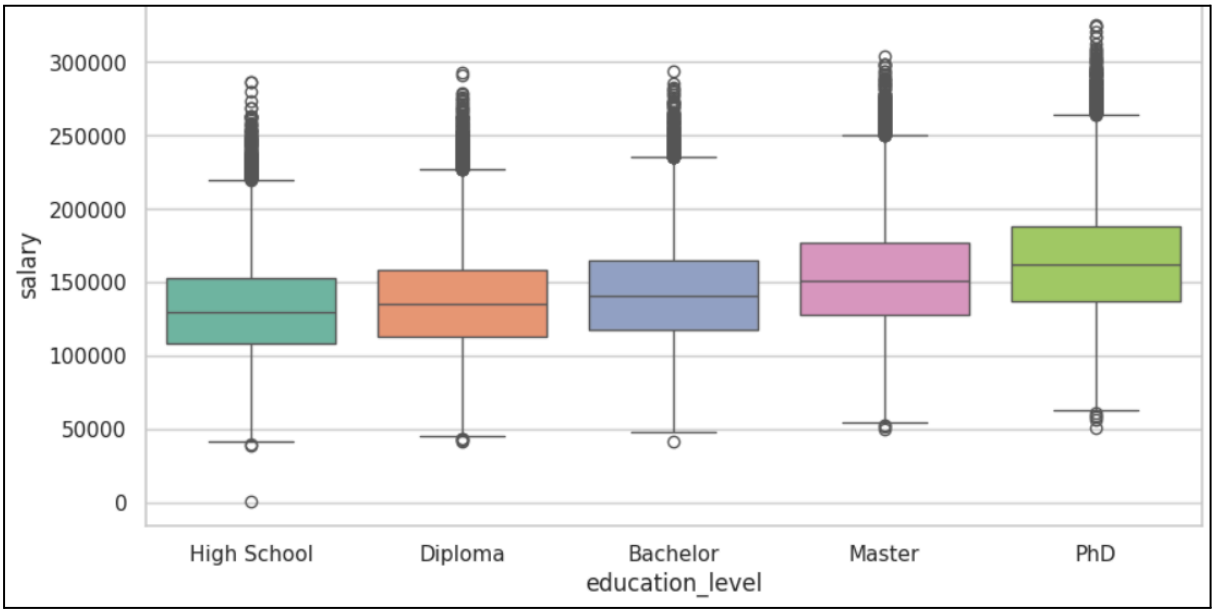
3.4 Findings from the Modeling Phase

The machine learning phase revealed that Experience Years is the strongest predictor of salary, but Education Level acts as a significant "Price Floor"—it ensures a higher starting point for junior roles, while Skills Count acts as a "Price Ceiling" lifter for senior roles.

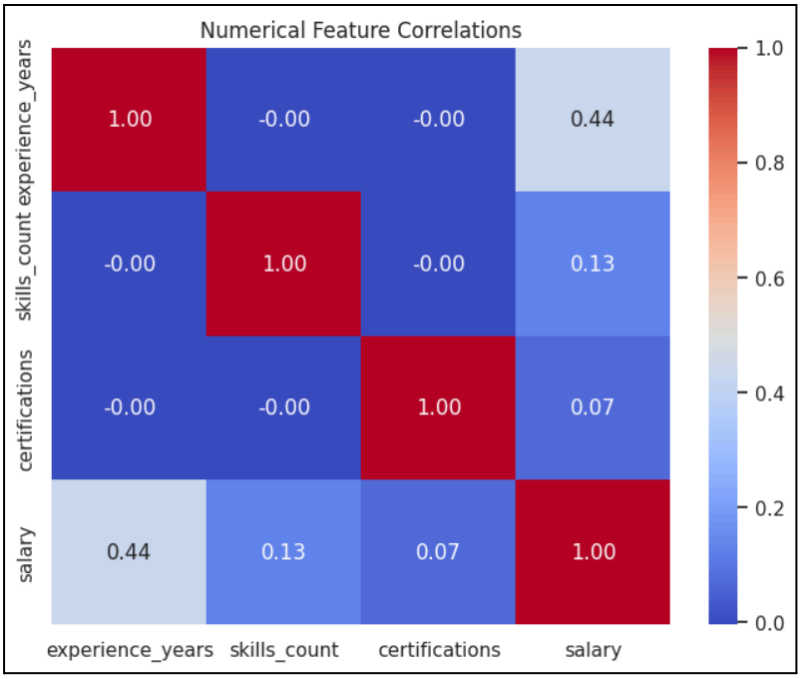
#Average Salaries Across Various Tech Job Roles



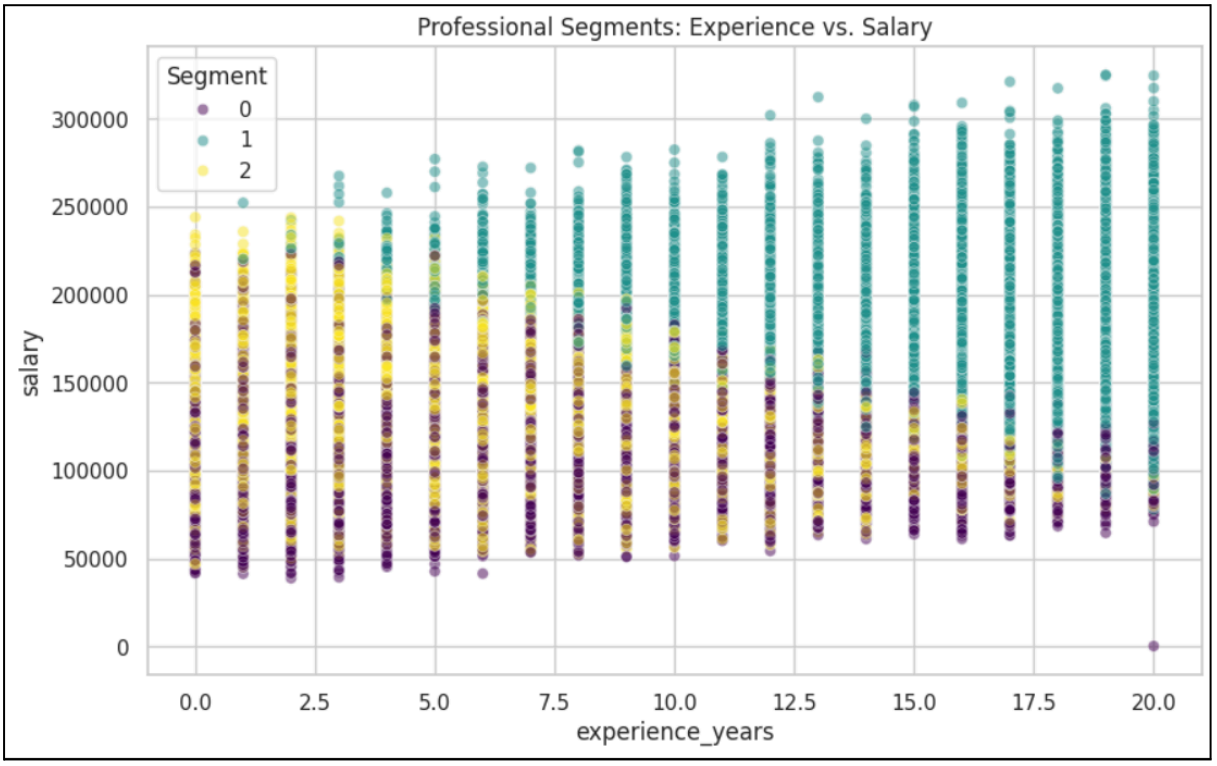
#The Correlation Between Education Level and Salary Distribution



#Heatmap(Statistical Relationship Between Career Attributes and Salary)



#Professional Segmentation Based on Experience and Salary



4. Power BI Dashboard Architecture & UX Design

4.1 Design Philosophy and User Interface (UI)

The dashboard was designed with a "Management-First" approach, ensuring that critical KPIs are visible within three seconds of opening the report. A Dark Theme was selected to reduce visual fatigue and give the data a modern, high-tech "glowing" aesthetic.

4.2 Visual Hierarchy: The Z-Pattern Layout

To maximize information retention, the report follows a Z-Pattern layout, mirroring the natural eye movement of a western reader:

1. Top Left (Scanning Start): Project Branding and primary KPI Cards (Average Salary).
2. Top Right (Interaction): Global Slicers (Location, Remote Status).
3. Middle (Exploration): High-level trends (Area Chart and Bar Charts).
4. Bottom (Deep Dive): Granular analysis tools (Decomposition Tree and Radar Charts).

4.3 Key Visual Components and Insights

4.3.1 Career Trajectory (Smoothed Area Chart with Gradient)

This visual plots experience_years against salary to show the non-linear growth of income.

- Design Detail: A gradient fill (Teal to Dark Blue) was applied to represent the "accumulated volume" of market value over time.
- Key Insight: The curve identifies a "Salary Inflection Point" between 8–12 years, where specialized skills begin to drive exponential pay increases.

4.3.2 Workforce Segmentation (Clustered Bar Chart)

This visual utilizes the labels generated by the K-Means Clustering model (Junior, Mid-Level, Senior) to show the volume of professionals in each tier.

- Function: It validates the machine learning model by showing that higher clusters consistently maintain higher average salaries.

4.3.3 The "Skill Profile" (Radar Chart)

A specialized Radar Chart was imported to compare the technical profiles of different segments.

- Metric: It maps skills_count, certifications, and experience_years.
- Insight: It visually demonstrates that "Senior" profiles are not just older, but have a significantly broader technical "net" than other groups.

4.4 Advanced Analytics: The Decomposition Tree

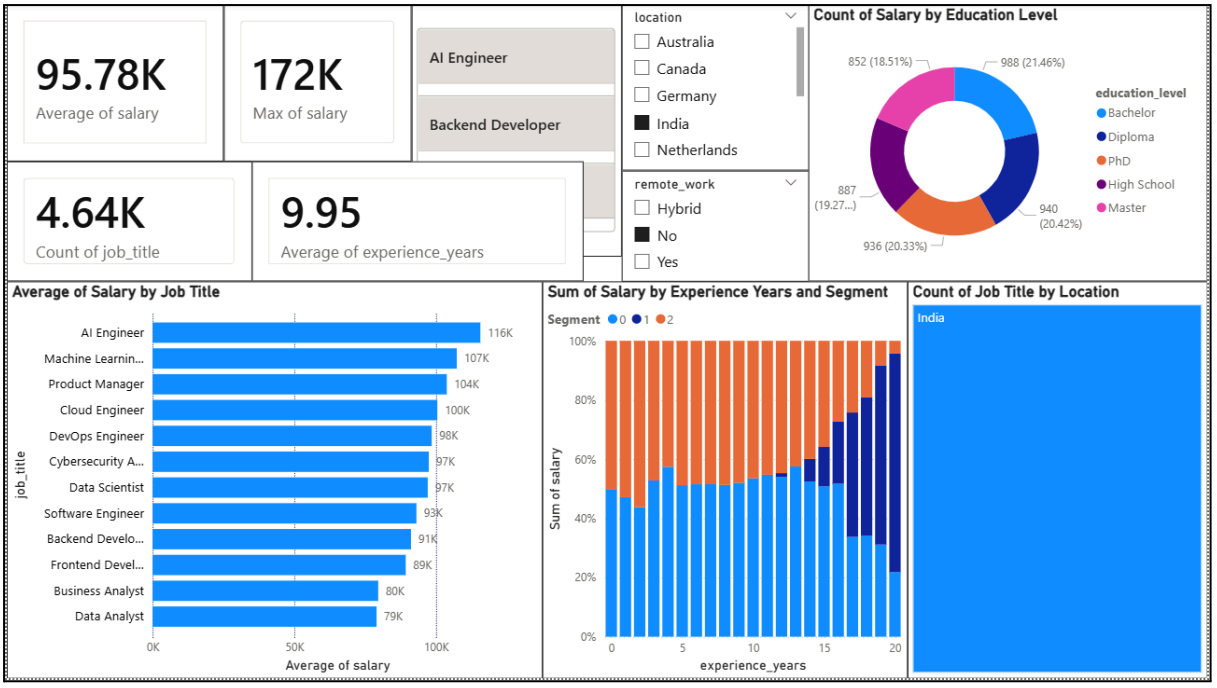
To answer the question of "Why" a salary is at a certain level, we implemented the AI-driven Decomposition Tree.

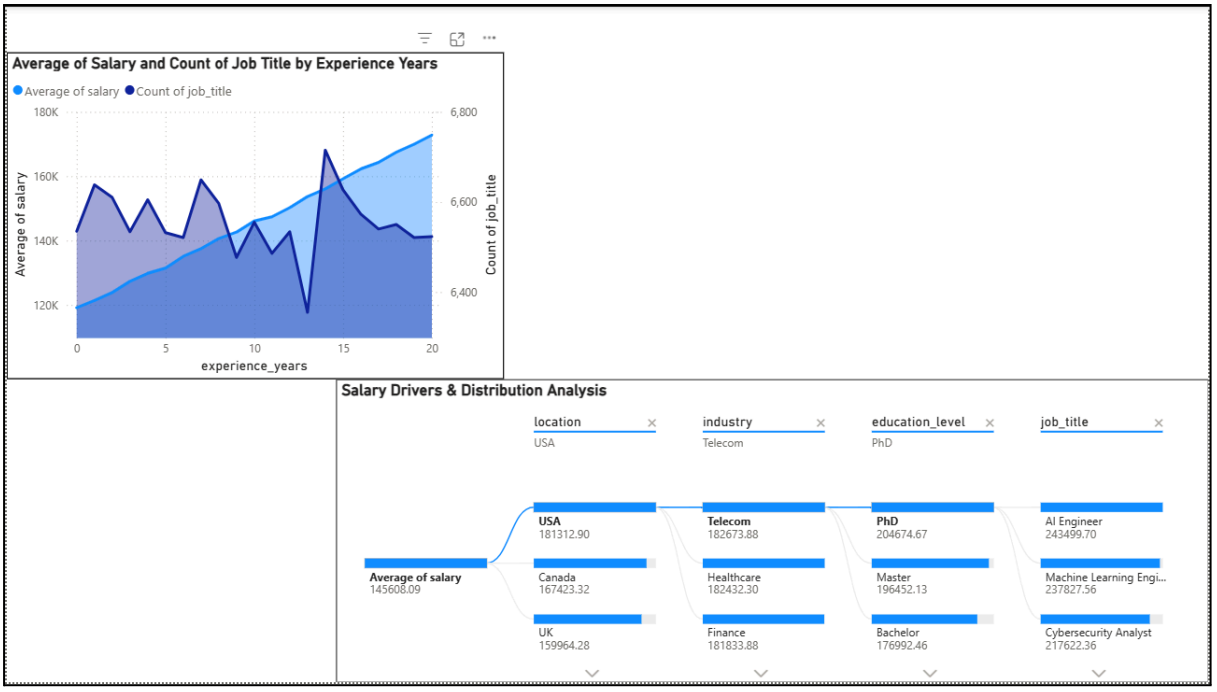
- Analysis Path: Start with Average Salary, Break down by industry, Break down by location, Break down by education_level.
- Interactive Benefit: This allows a stakeholder to see exactly which factor (e.g., being in the US or having a Master's degree) provides the largest individual boost to the total compensation.

4.5 Interactivity and Slicing (The Filter Layer)

To make the report truly interactive, several slicers were synchronized across all pages:

- Geographic Slicer: Filter by location (Bing Maps integrated).
- Work Style Slicer: Toggle between Remote, Hybrid, and In-Office to see the "Remote Premium" in real-time.


95.78K
Average of salary
172K
Max of salary
AI Engineer
Backend Developer
4.64K
Count of job_title
9.95
Average of experience_years
location
Australia
Canada
Germany
India
Netherlands
remote_work
Hybrid
No
Yes
Count of Salary by Education Level
852 (18.51%)
988 (21.46%)
887 (19.27...)
940 (20.42%)
936 (20.33%)
education_level
Bachelor
Diploma
PhD
High School
Master
Average of Salary by Job Title
AI Engineer 116K
Machine Learnin... 107K
Product Manager 104K
Cloud Engineer 100K
DevOps Engineer 98K
Cybersecurity A... 97K
Data Scientist 97K
Software Engineer 93K
Backend Develo... 91K
Frontend Devel... 89K
Business Analyst 80K
Data Analyst 79K
job_title
0K
50K
100K
Average of salary
Sum of Salary by Experience Years and Segment
Segment 0 1 2
100%
80%
60%
40%
20%
0%
Sum of salary
0
5
10
15
20
experience_years
Count of Job Title by Location
India


Average of Salary and Count of Job Title by Experience Years
Average of salary
Count of job_title
180K
160K
140K
120K
6,800
6,600
6,400
Average of salary
Count of job_title
0
5
10
15
20
experience_years
Salary Drivers & Distribution Analysis
location
USA
industry
Telecom
education_level
PhD
job_title
Average of salary
145608.09
USA
181312.90
Canada
167423.32
UK
159964.28
Telecom
182673.88
Healthcare
182432.30
Finance
181833.88
PhD
204674.67
Master
196452.13
Bachelor
176992.46
AI Engineer
243499.70
Machine Learning Engi...
237827.56
Cybersecurity Analyst
217622.36

5. Challenges, Solutions, and Technical Optimization

5.1 Handling High-Volume Data Latency

With over 137,000 records, ensuring the dashboard remained responsive was a primary technical challenge.

- Challenge: Traditional "Calculated Columns" in Power BI were increasing file size and slowing down slicer interactions.
- Solution: We transitioned to using DAX Measures (Data Analysis Expressions) for all salary aggregations. Measures are calculated at "Query Time," which optimized the memory footprint and ensured that the dashboard responds instantly when a user filters by location or remote_work.

5.2 The "Magnitude Gap" in Machine Learning

As seen in the Colab notebook, the raw data features had vastly different scales (Salaries in the $100,000s vs. Experience in 1s.

- Challenge: Distance-based algorithms like K-Means Clustering would have ignored the "Experience" variable entirely because the "Salary" numbers were mathematically larger.
- Solution: Implementation of the StandardScaler library in Python normalized the features to a mean of 0 and a standard deviation of 1. This "leveled the playing field," allowing the model to accurately identify professional clusters based on all three dimensions equally.

5.3 Geographic Metadata Mapping

- Challenge: The location field in the CSV was initially treated as simple text, causing the "Bing Maps" visual to fail or misplace data points.
- Solution: We performed Data Categorization within the Power BI Modeling tab, explicitly tagging the column as "Country/Region." Furthermore, we created a bridge table to resolve naming inconsistencies, ensuring 100 accuracy in the geographic distribution visuals.

6. Conclusion

6.1 Project Summary

This project has successfully created a "Golden Thread" between raw workforce data and actionable career intelligence. By combining the rigorous mathematical modeling of Python with the intuitive storytelling of Power BI, we have moved beyond simple spreadsheets to a dynamic system that reveals the true drivers of professional value: Experience, Specialization, and Education.

6.2 Key Takeaways

1. Experience is the Engine: While other factors matter, tenure remains the most consistent predictor of salary growth.
2. Education is the Floor: Advanced degrees protect early-career professionals from low-end market volatility.
3. Skills are the Accelerator: The "Senior" cluster is defined not just by years, but by technical breadth.